# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** \frac{4}{5} | $\sin\theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$ |
| 2 | **(b)** 7 m | $h = 7\tan 45^\circ = 7 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height = 7 m |
| 3 | **(a)** 4 cm | $PT^2 = OP^2 - OT^2 = 5^2 - 3^2 = 16$; $PT = 4$ cm |
| 4 | **(b)** 154 cm² | $\frac{22}{7} \times 7 \times 7$; Area = 154 cm² |
| 5 | **(c)** \frac{4}{3} | $\tan\theta = \frac{\text{perpendicular}}{\text{base}}$ |
| 6 | **(c)** $5\sqrt{3}$ m | $h = 5\tan 60^\circ = 5 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height = $5\sqrt{3}$ m |
| 7 | **(c)** 6 cm | $r^2 = OP^2 - PT^2 = 10^2 - 8^2 = 36$; $r = 6$ cm |
| 8 | **(c)** 44 cm | $2 \times \frac{22}{7} \times 7$; Circumference = 44 cm |
| 9 | **(a)** \frac{1}{2} | Standard value |
| 10 | **(a)** $3\sqrt{3}$ m | $h = 9\tan 30^\circ = 9 \times \frac{\sqrt{3}}{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height = $3\sqrt{3}$ m |
| 11 | **(d)** 25 cm | $OP^2 = r^2 + PT^2 = 7^2 + 24^2 = 625$; $OP = 25$ cm |
| 12 | **(b)** 7 cm | $r = \frac{44 \times 7}{2 \times 22}$; $r = 7$ cm |
| 13 | **(c)** 1 | Identity |
| 14 | **(d)** 9 m | $x = 9\cot 45^\circ$; Shadow = 9 m |
| 15 | **(d)** 14 cm | Perimeter = $PA + AO + OB + BP = 2(4) + 2(3)$; Perimeter = 14 cm |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | $\sin\theta = \frac{4}{5}$ (1) = $\frac{4}{5}$ (1). | 1+1 |
| 17 | Let the height be $h$; in the right triangle, $\tan 45^\circ = \frac{h}{10}$; $h = 10\tan 45^\circ = 10 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$) (1); Height = 10 m (1). | 1+1 |
| 18 | The radius to the point of contact $T$ is perpendicular to the tangent, so $\triangle OTP$ is right-angled at $T$; $PT^2 = OP^2 - OT^2 = 25^2 - 20^2 = 225$ (1); $PT = 15$ cm (1). | 1+1 |
| 19 | Area of a circle = $\pi r^2$; $\frac{22}{7} \times 3.5 \times 3.5$ (1); Area = 38.5 m² (1). | 1+1 |
| 20 | $\frac{1}{2} + \frac{1}{2}$ (1) = 1 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | $AC = \sqrt{3^2 + 4^2} = 5$ cm (1); $\sin A = \frac{BC}{AC} = \frac{4}{5}$, $\cos A = \frac{AB}{AC} = \frac{3}{5}$ (1); $\tan A = \frac{BC}{AB} = \frac{4}{3}$ (1). | 1+1+1 |
| 22 | Ladder is the hypotenuse; height = $12\sin 60^\circ$, base = $12\cos 60^\circ$ (1); $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$ (1); Height = $6\sqrt{3}$ m, distance = 6 m (1). | 1+1+1 |
| 23 | $PA = PB$ (tangents from an external point) and $OA = OB = r$ (1); Perimeter = $PA + AO + OB + BP = 2(16) + 2(12)$ (1); Perimeter = 56 cm (1). | 1+1+1 |
| 24 | Arc length = $\frac{\theta}{360} \times 2\pi r$ (1); $\frac{45}{360} \times 2 \times \frac{22}{7} \times 21$ (1); Arc = 16.5 cm (1). | 1+1+1 |
| | **OR:** Base = $\sqrt{5^2 - 4^2} = 3$ (1); $\cos A = \frac{3}{5}$ (1); $\tan A = \frac{4}{3}$ (1). | 1+1+1 |
| 25 | Angle of depression = angle of elevation of the top of the cliff from the boat = $60^\circ$ (1); $\tan 60^\circ = \frac{10\sqrt{3}}{x}$ so $x = 10\sqrt{3}\cot 60^\circ$ (1); Distance = 10 m (1). | 1+1+1 |
| | **OR:** $\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area = $2 \times \frac{1}{2} \times 3 \times 4$ (1); Area = 12 cm² (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Hypotenuse = $\sqrt{4^2 + 3^2} = 5$ (1); $\sin\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$ (1); $\sec\theta = \frac{5}{3}$ (1). | 1+1+1 |

| | | |
|---|---|---|
| | **OR:** Distances of the points from the foot: $h\cot 30^\circ = 18$ m and $h\cot 60^\circ = 6$ m (2); The points are on opposite sides, so the distances add (2); Distance $= 24$ m (1). | 2+2+1 |
| **27** | $OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{5^2 - 3^2} = 4$ cm; perimeter $= 2(4+3)$ (2); Perimeter $14$ cm; area $= 2 \times \frac{1}{2} \times 3 \times 4 = 12$ cm$^2$ (1). | 2+2+1 |
| | **OR:** Outer radius $= 21$ m (2); Area $= \pi(R^2 - r^2) = \frac{22}{7}(21^2 - 14^2)$ (2); Area of path $= 770$ m$^2$ (1). | 2+2+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 2

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 7 cm | $r^2 = \frac{154\times7}{22} = 49$; $r = 7$ cm |
| 2 | **(a)** 65^\circ | $\sin\theta = \cos(90^\circ - \theta)$ |
| 3 | **(c)** 12 m | $\tan 45^\circ = \frac{12}{x}$ so $x = 12\cot 45^\circ$; Distance $= 12$ m |
| 4 | **(c)** $70^\circ$ | In quadrilateral $OAPB$: $\angle APB = 360^\circ - 90^\circ - 90^\circ - 110^\circ$; $\angle APB = 70^\circ$ |
| 5 | **(c)** 231 cm² | $\frac{60}{360} \times \frac{22}{7} \times 21^2$; Area $= 231$ cm$^2$ |
| 6 | **(b)** $\cos\theta$ | $\sec\theta = 1/\cos\theta$ |
| 7 | **(d)** $4\sqrt{3}$ m | Height $= 8\sin 60^\circ = 8 \times \frac{\sqrt{3}}{2}$; Height reached on the wall $= 4\sqrt{3}$ m |
| 8 | **(d)** $60^\circ$ | $\angle AOP = 180^\circ - 90^\circ - 30^\circ$; $\angle AOP = 60^\circ$ |
| 9 | **(d)** 22 cm | $\frac{60}{360} \times 2 \times \frac{22}{7} \times 21$; Arc $= 22$ cm |
| 10 | **(c)** \frac{12}{13} | $\sin\theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$ |
| 11 | **(a)** line of sight | Definition |
| 12 | **(c)** one point | Definition of a tangent |
| 13 | **(d)** 36 cm | Arc $= \pi r = 22$ cm; diameter $= 14$ cm; Perimeter $= 36$ cm |
| 14 | **(b)** \frac{12}{5} | $\tan\theta = \frac{\text{perpendicular}}{\text{base}}$ |
| 15 | **(d)** 12 m | $h = 12\tan 45^\circ = 12 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 12$ m |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Let the shadow be $x$; $\tan 45^\circ = \frac{14}{x}$; $x = 14\cot 45^\circ$ (1); Shadow $= 14$ m (1). | 1+1 |
| 17 | Radius $\perp$ tangent, so $OP$ is the hypotenuse of a right triangle with sides $r$ and $PT$; $OP^2 = r^2 + PT^2 = 4^2 + 3^2 = 25$ (1); $OP = 5$ cm (1). | 1+1 |
| 18 | Circumference $= 2\pi r$; $2 \times \frac{22}{7} \times 3.5$ (1); Circumference $= 22$ m (1). | 1+1 |
| 19 | Using $\sin^2\theta + \cos^2\theta = 1$ (1); value $= 1$ (1). | 1+1 |
| 20 | Kite string is the hypotenuse of the right triangle; $h = 20\sin 30^\circ = 20 \times \frac{1}{2}$ (1); Height $= 10$ m (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Perimeter $=$ arc $+ 2r$ (1); Arc $= \frac{60}{360} \times 2\pi \times 21 = 22$ cm (1); Perimeter $= 22 + 42 = 64$ cm (1). | 1+1+1 |
| 22 | Use $\sin^2 30^\circ = \frac{1}{4}$, $\cos^2 60^\circ = \frac{1}{4}$ (2); value $= \frac{1}{2}$ (1). | 2+1 |
| 23 | Let the shadow be $x$; $\tan 60^\circ = \frac{12\sqrt{3}}{x}$ (1); $x = 12\sqrt{3}\cot 60^\circ$ (1); Shadow $= 12$ m (1). | 1+1+1 |
| 24 | $OA \perp PA$, so $\triangle OAP$ is right-angled at $A$ (1); $\angle AOP = 180^\circ - 90^\circ - 20^\circ$ (1); $\angle AOP = 70^\circ$ (1). | 1+1+1 |
| | **OR:** Area of semicircle $= \frac{1}{2}\pi r^2$ (1); $\frac{1}{2} \times \frac{22}{7} \times 7^2$ (1); Area $= 77$ cm$^2$ (1). | 1+1+1 |
| 25 | $AC = \sqrt{5^2 + 12^2} = 13$ cm (1); $\sin A = \frac{BC}{AC} = \frac{12}{13}$, $\cos A = \frac{AB}{AC} = \frac{5}{13}$ (1); $\tan A = \frac{BC}{AB} = \frac{12}{5}$ (1). | 1+1+1 |
| | **OR:** Ladder is the hypotenuse; height $= 16\sin 60^\circ$, base $= 16\cos 60^\circ$ (1); $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$ (1); Height $= 8\sqrt{3}$ m, distance $= 8$ m (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | $PR = 10$ cm (1); $\sin P = \frac{4}{5}$, $\cos P = \frac{3}{5}$, $\tan P = \frac{4}{3}$ (2); $\left(\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2 = 1 = 1$ (2). | 1+2+2 |
| | **OR:** Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 60^\circ$, which differ by 6 m (2); $h(\cot 30^\circ - \cot 60^\circ) = 6$ gives $h = 6 \div (\frac{2\sqrt{3}}{3})$ (2); Height $= 3\sqrt{3}$ m (distances 9 m and 3 m) (1). | 2+2+1 |

| | | |
|---|---|---|
| **27** | The chord touches the smaller circle at $M$, so $OM \perp$ chord and $M$ is the midpoint of the chord (2); Half-chord $= \sqrt{25^2 - 7^2} = 24$ cm (2); Chord $= 2 \times 24 = 48$ cm (1). | 2+2+1 |
| | **OR:** The largest circle has diameter equal to the side, so $r = \frac{s}{2}$ (2); Circle area $= \frac{22}{7} \times 7^2 = 154$ cm$^2$; square $= 196$ cm$^2$ (2); Left over $= 42$ cm$^2$ (1). | 2+2+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 3

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 8 cm | $PT^2 = OP^2 - OT^2 = 10^2 - 6^2 = 64$; $PT = 8$ cm |
| 2 | **(d)** 25 cm | Quarter arc $= \frac{\pi r}{2} = 11$ cm; $2r = 14$ cm; Perimeter $= 25$ cm |
| 3 | **(c)** \frac{1}{2} | Standard value |
| 4 | **(a)** $8\sqrt{3}$ m | $h = 8\tan 60^\circ = 8 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 8\sqrt{3}$ m |
| 5 | **(c)** 5 cm | $r^2 = OP^2 - PT^2 = 13^2 - 12^2 = 25$; $r = 5$ cm |
| 6 | **(d)** 616 cm² | $\frac{22}{7} \times 14 \times 14$; Area $= 616$ cm$^2$ |
| 7 | **(c)** 1 | Identity |
| 8 | **(b)** $4\sqrt{3}$ m | $h = 12\tan 30^\circ = 12 \times \frac{\sqrt{3}}{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 4\sqrt{3}$ m |
| 9 | **(d)** 25 cm | $OP^2 = r^2 + PT^2 = 15^2 + 20^2 = 625$; $OP = 25$ cm |
| 10 | **(d)** 88 cm | $2 \times \frac{22}{7} \times 14$; Circumference $= 88$ cm |
| 11 | **(b)** 50^\circ | $\sin\theta = \cos(90^\circ - \theta)$ |
| 12 | **(a)** 15 m | $x = 15\cot 45^\circ$; Shadow $= 15$ m |
| 13 | **(a)** 28 cm | Perimeter $= PA + AO + OB + BP = 2(8) + 2(6)$; Perimeter $= 28$ cm |
| 14 | **(c)** 14 cm | $r = \frac{88 \times 7}{2 \times 22}$; $r = 14$ cm |
| 15 | **(d)** \frac{5}{3} | Reciprocal |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | $OA \perp PA$ and $OB \perp PB$, so $\angle OAP = \angle OBP = 90^\circ$; In quadrilateral $OAPB$: $\angle APB = 360^\circ - 90^\circ - 90^\circ - 120^\circ$ (1); $\angle APB = 60^\circ$ (1). | 1+1 |
| 17 | Area of sector $= \frac{\theta}{360} \times \pi r^2$; $\frac{45}{360} \times \frac{22}{7} \times 21^2$ (1); Area $= 173.25$ cm$^2$ (1). | 1+1 |
| 18 | $\sin\theta = \frac{12}{13}$ (1) $= \frac{12}{13}$ (1). | 1+1 |
| 19 | Let the height be $h$; in the right triangle, $\tan 45^\circ = \frac{h}{16}$; $h = 16\tan 45^\circ = 16 \times 1$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$) (1); Height $= 16$ m (1). | 1+1 |
| 20 | The radius to the point of contact $T$ is perpendicular to the tangent, so $\triangle OTP$ is right-angled at $T$; $PT^2 = OP^2 - OT^2 = 26^2 - 24^2 = 100$ (1); $PT = 10$ cm (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | $PA = PB$ (tangents from an external point) and $OA = OB = r$ (1); Perimeter $= PA + AO + OB + BP = 2(24) + 2(7)$ (1); Perimeter $= 62$ cm (1). | 1+1+1 |
| 22 | Arc length $= \frac{\theta}{360} \times 2\pi r$ (1); $\frac{72}{360} \times 2 \times \frac{22}{7} \times 21$ (1); Arc $= 26.4$ cm (1). | 1+1+1 |
| 23 | Base $= \sqrt{13^2 - 12^2} = 5$ (1); $\cos A = \frac{5}{13}$ (1); $\tan A = \frac{12}{5}$ (1). | 1+1+1 |
| 24 | Angle of depression = angle of elevation of the top of the cliff from the boat $= 60^\circ$ (1); $\tan 60^\circ = \frac{15\sqrt{3}}{x}$ so $x = 15\sqrt{3}\cot 60^\circ$ (1); Distance $= 15$ m (1). | 1+1+1 |
| | **OR:** $\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 6 \times 8$ (1); Area $= 48$ cm$^2$ (1). | 1+1+1 |
| 25 | Perimeter = arc $+ 2r$ (1); Arc $= \frac{90}{360} \times 2\pi \times 21 = 33$ cm (1); Perimeter $= 33 + 42 = 75$ cm (1). | 1+1+1 |
| | **OR:** Use $\tan^2 45^\circ = 1,\ \sin^2 90^\circ = 1$ (2); value $= 2$ (1). | 2+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Hypotenuse $= \sqrt{12^2 + 5^2} = 13$ (1); $\sin\theta = \frac{12}{13}$, $\cos\theta = \frac{5}{13}$ (1); $\sec\theta = \frac{13}{5}$ (1). | 1+1+1 |

| | | |
|---|---|---|
| | **OR:** Distances of the points from the foot: $h\cot 30^\circ = 27$ m and $h\cot 60^\circ = 9$ m (2); The points are on opposite sides, so the distances add (2); Distance $= 36$ m (1). | 2+2+1 |
| **27** | $OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{10^2 - 6^2} = 8$ cm; perimeter $= 2(8+6)$ (2); Perimeter $28$ cm; area $= 2 \times \frac{1}{2} \times 6 \times 8 = 48$ cm$^2$ (1). | 2+2+1 |
| | **OR:** Outer radius $= 28$ m (2); Area $= \pi(R^2 - r^2) = \frac{22}{7}(28^2 - 21^2)$ (2); Area of path $= 1078$ m$^2$ (1). | 2+2+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND